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Incels and Extremism

Digital Methods: Exam

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1 Introduction (All)

Since the early 2010s, Western countries have experienced a wave of mass violence motivated by a contemporary form of misogyny. An illustrative example is the Toronto van attack committed by 25-year-old Alek Minassian in April, 2018. Minutes before driving his van into a crowded sidewalk, killing 10 and injuring 16, Minassian posted on Facebook: *"The Incel Rebellion has already begun! We will overthrow all the Chads and Stacys! All hail the Supreme Gentleman Elliot Rodger!"* (Barlow, 2018). The term incels, a portmanteau for involuntary celibates, refers to an online community of more than 60,000 predominantly young, white, and Western men. Spread across multiple forums, incels contemplate their inability to obtain sex, arguing that they are overlooked by 'Stacys' in favor of 'Chads' - community slang for physically attractive women and men (O'Donnell & Shor, 2022). Elliot Rodger is an earlier perpetrator of misogynistically motivated mass violence who wrote and distributed a manifesto, leading to widespread idolization in the online incel community. As is evident from Minassian's actions and accompanying Facebook post, incel culture has led to public display of statements that glorify violence and individual actions that must be categorized as terrorism.

In spite of this, it is a subject of scholarly debate whether the incel community writ large is extremist (O'Hanlon et al., 2024). This lack of consensus is at least in part attributable to disagreements in the literature over what the concept of extremism entails. Extremism is often bundled together with other contested concepts such as radicalization or terrorism, and conceptual approaches to extremism have historically focused on political and religious belief systems operating in institutionalized settings such as states or global terrorist organizations (Bötticher, 2017). Obtaining an improved understanding of the incel worldview is important in itself for the study of toxic social dynamics online and for the development of prevention policy. But incel culture is also a deviant case of a community that engages in ideologically motivated mass violence in the sense that it is decentralized and its defining issue is that of gender rather than religion or politics in a conventional sense. Therefore, an in-depth study of the belief systems of incels may offer novel analytical insights into to the concept of extremism itself, thereby contributing to conceptual debates in the literature. In this assignment we thus address a two-fold research question. First, we seek to answer the following question: *What tenets in incel culture resemble the structure of extremist ideology?* Second, in doing so, we design our research so as to cultivate the discovery of relevant

and surprising observations that allow us to learn something about the concept of extremism itself.

To study incel culture as a case of extremism, we need an elaborate theoretical framework. The next section in this assignment therefore presents a theory of the structure of extremist ideology based on contemporary work in the fields of terrorism studies and philosophy. To study the belief systems of incels, we employ a digital mixed methods approach that uses computational text analysis tools to assist human reading and interpretation. The third section elaborates on this approach and describes the site of our study - the *incels.is* forum. The fourth section presents qualitative and quantitative results pertaining to discourses displayed in the forum. In the final section, we expand the scope and ask whether we can learn something about extremism from our study of the belief systems of incels.

2 Theory: The Structure of Extremist Ideology

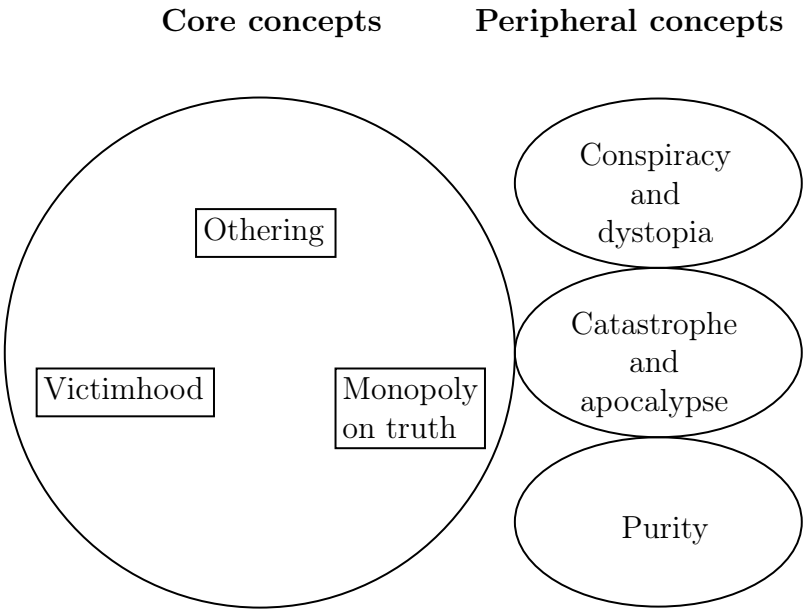
This section provides a theoretical account of extremist ideology. An ideology is a framework for making sense of the world, and although extremist ideologies differ widely in their contents, they share a structure (Berger, 2018, pp. 43–44; Cassam, 2021, pp. 13–14). To develop an account of this shared structure, we argue that extremist ideologies have core and peripheral concepts. The core concepts are the necessary components of an extremist ideology that attain meaning and prescribe action through their relation to a range of contingent peripheral concepts (Cassam, 2021, pp. 111–112). Whereas the core concepts in extremist ideology pertain to the nature of collective identities, the peripheral concepts specify a range of causes and consequences of the relationship between these identities. The combination of the core concepts with one or more of the peripheral concepts creates a coherent worldview that often justifies the use of violence. Figure 1 provides an overview of the structure of extremist ideology.

2.1 Core Concepts

2.1.1 Othering (Noel)

According to social identity theory, the formation of collective identities is a process that revolves around the construction of an in-group and an out-group (Tajfel & Turner, 1979). Collective identities are defined by in-group narratives of shared beliefs, traits,

Figure 1: The Structure of Extremist Ideology



and practices. The demarcation and differentiation of the in-group identity requires its juxtaposition to the out-group identity. In other words, defining 'us' requires a definition of 'them', where 'us' and 'them' must be different (Berger, 2018, pp. 54–57). While the formation of collective identities is a universal social process, it has two special features in the context of extremist ideology. First, in most settings, group membership is context-dependent, changes over time, and is a matter of subjective judgment. In contrast, a central element of extremist ideology is its absolutist conception of the nature of collective identities. For example, the Nazi regime in Germany during the 1930s and 1940s crafted detailed and restrictive laws about heritage and ethnicity in an effort to erase any doubt about who could be considered German and who could be considered Jewish (Berger, 2018, pp. 32, 51–54). Second, in the context of extremist ideology, a central aspect of collective identity formation is othering - the attribution of superiority to the in-group and inferiority to the out-group. As such, extremism is characterized by its view of collective identities as exclusive and absolute categories that are organized in a hierarchy (Cassam, 2021, pp. 104–106; Lumsden & Harmer, 2019, pp. 13–15).

2.1.2 Victimhood (Signe)

A second core element of extremist ideology is the conviction that members of the in-group are the victims of persecution, injustice, and humiliation perpetrated by members of the out-group. The extremist conception of victimhood is closely tied to the collective identity of the out-group which is conceived of as intrinsically harmful to the in-group. Accordingly, extremists' understanding of their own victimhood is grounded in imagined, exaggerated, or misattributed persecution or humiliation. The close connection between the out-group identity and the victimization of the in-group is often established in extremist discourse by situating the groups in a struggle that is historically scaled or determined by greater powers such as divinity or nature (Berger, 2018, pp. 75–76; Cassam, 2021, pp. 95–96; Karell & Freedman, 2019).

2.1.3 Monopoly on Truth (Sarah)

The final core component of extremist ideology is its commitment to a comprehensive and uncontestable ontology. Extremists claim a monopoly on truth from which they can derive final solutions to the central problems of society. This position has two consequences. First, because extremists believe that they are the proponents of the one uniquely right way of living, they are anti-pluralists seeking to bring about comprehensive homogenization of society. Second, because extremists believe their worldview to be the one good and true vision for society, they are dogmatic and unwilling to engage in any form of compromise (Cassam, 2021, pp. 100–104, 106–107; Bötticher, 2017).

2.2 Peripheral Concepts

2.2.1 Conspiracy and Dystopia (Stinus)

The first peripheral element of extremist ideology is its tendency to develop conspiracy theories and use them in dystopian narratives. Conspiracy theories argue that the out-group controls the success or survival of the in-group through secretive means that must be exposed. Such theories allow the in-group to develop a detailed and cohesive explanation for their imagined persecution and humiliation by the out-group (Berger, 2018, pp. 84–89; Cassam, 2021, pp. 109–110). In the context of extremist ideology, conspiracy theories are sometimes part of a larger dystopian narrative of a fundamentally inhumane society governed by a regime that favors the out-group and disadvantages the in-group (Berger, 2018, pp. 89–91).

2.2.2 Catastrophe and Apocalypse (Stinus)

Extremists often engage in two related but different ways of foretelling great disaster. First, catastrophic thinking revolves around an impending catastrophe with consequences for the in-group exclusively. For example, the Islamist terrorist organization al Qaeda issued public warnings in the early 2000s that Europeans and Americans were soon to commit a comprehensive and global genocide on Muslims and Arabs (Berger, 2018, p. 92; Cassam, 2021, p. 110). Second, apocalyptic thinking projects the end of history following a conflict between the in-group and the out-group that will either in itself or through divine intervention unleash a cosmic wave of destruction (Berger, 2018, pp. 92–94).

2.2.3 Purity (Noel)

The final peripheral component of extremist ideology is its preoccupation with purity. In this context, purity refers to the correspondence between a narratively constructed archetypical in-group identity and the actual beliefs, traits, and practices of the current members of the in-group. An obsession with purity is often reflected in extremist discourse through a moralizing and intimate rhetoric focusing on the ways of life of the members of the in-group. Impurity becomes a crisis when the members of the in-group start to resemble the members of the out-group through a process of infiltration (Berger, 2018, pp. 83–84; Cassam, 2021, pp. 91–94; Karell & Freedman, 2019).

3 Data and Methods

3.1 Data Site: *incels.is*

As our data site to study incel culture, we used the website *incels.is*. *Incels.is* is an online community *"for men that struggle with[...]romantic relationships with women despite trying"*, welcoming all men *"as long as [they] are an incel"*. The site's main content is user-initiated threads posted in five forums: "Must-Read Content", "Inceldom Discussion", "Offtopic", "The Sewers", and "Ban Appeals". When users post in the forums, they assign the post a so-called prefix, which is a predefined topic label or category. One must sign up to post on the site. Below, we discuss how we followed the investigative operations outlined by Kozinets (2020) in the process of data collection.

3.1.1 Initial Investigation: Simplification, Search and Scouting (Signe)

The first task when conducting netnography is to locate relevant data sites. This includes *simplification* which means rendering key elements of our research question into searchable terms (Kozinets, 2020, pp. 216–218). As we wish to study the expression of forms of extremism in the incel community as a whole, we are looking for data sites that plausibly represent the incel community. While the incel community is not a bounded or asserted group or organization, we know that the term "incel" is used by members of the community to self-identify. Searching on specific, well-known platforms such as Reddit is an obvious approach to *scouting*. However, several of the biggest subreddits have been banned in the past years due to harmful content. Therefore, we *searched* for forums on Google search using the specific term "incel" followed by "forum" and "discussion". We found mentions of the site *incels.is* - Involuntary Celibate Forum (Kozinets, 2020, pp. 219–226; Gillet & Suzor, 2022). Two immediately appealing features of the site are its large size of 25,000 members and the site's clear affiliation to inceldom.

3.1.2 Selecting: Five Criteria for Selecting the Data Site (Signe)

In *selecting* our data and data site, we use Kozinets' five criteria: *relevance, activity, interactivity, diversity, and richness*. Firstly, the site *incels.is* is *relevant* because it is highly related to our research question (Kozinets, 2020, p. 227). We carried out some preliminary immersion into the site to ensure that there would be relevant discussions. Here, we found that "Inceldom Discussion" is the only forum directly related to inceldom, described as a forum to *"talk about inceldom, coping strategies, and anything else related to involuntary celibacy"*. The other forums regard discussions about lifestyle, music and meta discussions about the site. Therefore, we limit our data collection to "Inceldom Discussion".

Secondly, the data site is *active* as there are several posts per hour. Thirdly, the data site is *interactive*. Users can initiate threads in which other users can post. This criterion is also the reason why we limit our data collection to the "Inceldom Discussion" forum to ensure interaction. This multi-way communication allows for both interaction, emotion, and *"the potential for communal types of relations"* (Kozinets, 2020, p. 228)

Fourthly, the criterion of *diversity* is essential as it prevents one-sided data that might not reflect the empirical reality under investigation properly (Kozinets, 2020, p. 229). While we immediately noticed that some categories, such as "ToxicFeminity", indicate the existence of shared beliefs - here, beliefs about women's fault in male

injustice - our preliminary immersions also revealed disagreements about beliefs and appropriate behavior. This is essential to capture the range of beliefs and emotions that are prevalent in this community.

Lastly, the criterion of *richness* entails that the data site contains a certain amount of rich data (Kozinets, 2020, p. 229). Often, rich data take the form of longer, textual descriptions that provide context, background story, or reflections that allow netnographers to analyze and interpret text in a deeper manner. Through our preliminary immersion, we found several examples of such rich data entries.

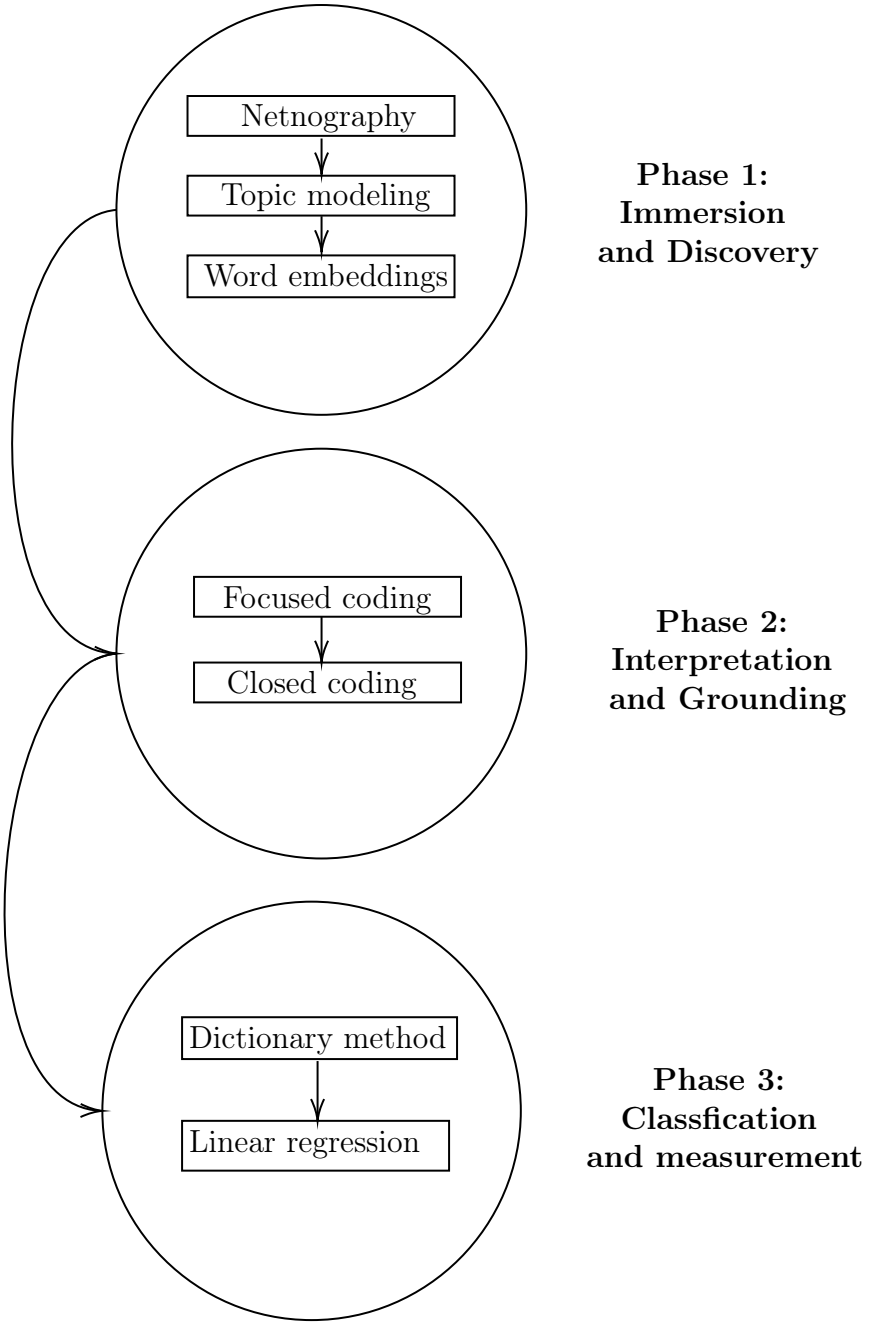
3.1.3 Saving the Data: Scraping (Signe)

To *save* the data, we scraped the "Inceldom Discussion" forum part of the site (Kozinets, 2020, p. 236). We collected all threads, saving the initial posts and comments separately. Additionally, we scraped the number of replies, views, topic labels, and usernames. After removing duplicates and observations with missing values, our dataset contains $N = 268,923$ observations.

3.2 Analysis Design: Computer-Assisted Learning and Measurement

To study incel culture, we use a digital mixed methods approach that follows the computer-assisted learning and measurement (CALM) framework with adjustments and additions. CALM is a framework for quali-quant content analysis in which the justification for interpretation and classification of texts relies on qualitative human reading rather than computational models (Carlsen & Ralund, 2022). The framework follows the structure and logic of an exploratory sequential mixed methods analysis by moving through three interdependent and complementary phases: immersion and discovery, interpretation and grounding, and classification and measurement. This section describes the methods used in each of the phases and the meta-analytical procedures used to connect them into an integrated analysis of extremist rhetoric in the incel community (Small, 2011; Frederiksen et al., 2023, pp. 137–143). Figure 2 provides an overview of the analysis design.

Figure 2: Analysis Design



3.2.1 Immersion and Discovery

The goal of the first phase in our analysis design is to discover discursive patterns in incel culture and to develop interpretative competence. To achieve this, we sequentially undertake three different forms of exploratory analysis. First, we engage in netnographic immersion in the *incels.is* forum to acquire an initial understanding of the language used and the themes discussed. Second, we run a topic model to consolidate the discursive patterns discovered through netnography and to arrive at a list of seed words for each theme. Finally, we use a word embedding model to expand the lists of seed words. Upon completion of this first immersion and discovery phase, we will have obtained a list of analytical themes with associated keywords that we can use for retrieving a sample of documents for close reading in the second interpretation and grounding phase. Below we describe each of the analytical operations of the immersion and discovery phase.

3.2.1.1 Netnography (Stinus) Our very first point of entrance to the study of incel culture is netnographic immersion in the *incels.is* forum. We employ four types of immersive data operations. First, we *reconnoitre* the forum to develop an understanding of its structure, how the users engage with its features, and to search for threads with analytical potential (Kozinets, 2020, pp. 286–290). To ensure topical breadth in our immersive operations, we repurpose the forum's native filtering function and separately search for threads that are labelled with each of the forum's 21 prefixes. Second, we *record* our immersive data operations in an immersion journal submitted as supplementary material along with this assignment on *eksamen.ku.dk*. For each thread we read, we write up a short description of its content, reflect on its meaning through the analytical lens of extremism, and label it with one or more analytical themes. As such, writing up the journal also comprises the third and fourth immersive data operations - *researching* and *reflecting* (Kozinets, 2020, pp. 290–302).

Netnography plays multiple important roles in the immersion and discovery phase and for the integrated analysis as a whole. When writing up the immersion journal, we engage in a semi-open coding practice that takes the theoretical structure of extremist ideology as point of departure but remains open to relevant but surprising themes. This practice contributes to defamiliarizing the concept of extremism in the context of incel culture, thereby cultivating surprising observations that we can potentially leverage for drawing abductive inferences (Timmermans & Tavory, 2022, pp. 72–73). Netnography thus contributes to developing the analytical foci that we carry forward in the second

interpretation and grounding phase of the analysis. At the same time, we acquire an interpretative understanding of the language used in the incel community through the immersive data operations. This understanding helps us to interpret the output of computational models and assists our qualitative analysis. As such, netnographic immersion is what allows us to attune our operationalizations to the practices of the users of *incels.is* forum through the analytical adoption of emic rather than etic language (Venturini et al., 2018; Grigoropoulou & Small, 2022; Carlsen & Ralund, 2022). Finally, the immersive data operations leave us with preliminary lists of seed words relating to each of the tentative analytical foci. We consolidate the analytical foci and expand upon the lists of seed words using two different computational models. The next subsections describe these models and their usage in the context of our analysis in turn.

3.2.1.2 Topic Modeling (Stinus) Topic models are a class of unsupervised learning algorithms that assign documents in a corpus to multiple latent topics based on co-occurrences of words (Nelson, 2020; Grimmer et al., 2022, pp. 147–149; Blei, 2012). However, a true interpretative understanding of the meaning of a text cannot be claimed without an understanding of the context of the situation that produced the text, which a topic model does not provide (Carlsen & Ralund, 2022).

We estimate the most canonical topic model, Latent Dirichlet Allocation (LDA), with twenty topics on our corpus and use it to assist our qualitatively driven exploration of discursive patterns in the *incels.is* forum. While netnography is our primary method for discovering analytical themes in the forum, LDA allows us to take an exploratory view that, unlike netnography, covers the corpus broadly. Hence, we use LDA to ensure ourselves against selective and biased exploration, thus corroborating the tentative analytical foci developed through netnographic immersion. At the same time, the process of interpretating the output of the LDA model forces us to consider unexpected word co-occurrence patterns, thus contributing to defamiliarizing the concept of extremism similarly to open coding, albeit on a larger scale (Karell & Freedman, 2019). Finally, we treat the output of the LDA model as suggestions for additions and adjustments to our lists of seed words developed through netnographic immersion.

3.2.1.3 Word Embeddings (Signe) We train word embeddings using Word2Vec with skipgram architecture and 300 dimensions on our corpus to expand our seed words. Word embeddings embed words in a corpus in a shared multi-dimensional vector space. Assuming that meaning is derived from context, proximity of vectors in the space

reflects semantic similarity of words. This makes Word2Vec a useful tool for keyword expansion (Rodriguez & Spirling, 2022; Grimmer et al., 2022, pp. 78–84; Carlsen & Ralund, 2022; Blok et al., 2022). The expansion operation leaves us with a keyword list for each analytical theme. We use these keyword lists to query our data. From the resulting query set, we randomly sample 180 threads. We split this random sample in to a close reading set of 120 threads and training set of 60 threads that each serve specific purposes in the second and third phase of the analysis design.

3.2.2 Interpretation and Grounding

The purpose of the second phase in our analysis design is to develop an analytical argument about the discourses displayed in the *incels.is* forum through the theoretical lens of the structural properties of extremist ideology. To do so, we engage in focused coding of our close reading set of 120 threads. Apart from the development of an argument about incel ideology, focused coding also allows us to construct a codebook and a dictionary. We use the codebook for closed coding of our test set of 60 threads and carry the annotated test set and the dictionary forward for classification and validation purposes in the third phase of the analysis design. Here we describe the focused and closed coding practices of the interpretation and grounding phase.

3.2.2.1 Focused Coding and Dictionary Development (Noel) Focused coding is the main vehicle of our analysis design for making sense of incel ideology. Focused coding entails critically and systematically reading through all the 120 threads in our close-reading set in a three-step process. First, we identify an index case for each of the analytical foci carried forward from the immersion and discovery phase. The index cases are the threads that most clearly capture each conceptual aspect of extremist ideology as it unfolds in incel culture (Timmermans & Tavory, 2022, pp. 92–93). Second, we position all threads relating to an analytical theme against the index case in an effort to uncover variation within each theme and investigate connections between themes. Finally, through the process of focused coding, we develop an analytical argument about the structure of incel ideology and how it relates to theoretical extremism (Timmermans & Tavory, 2022, pp. 94–96). As a biproduct of the focused coding process, we construct a dictionary that maps analytical themes to lists of words. We use this dictionary as a classification device in the third phase of our analysis design.

3.2.2.2 Codebook Development and Closed Coding (Stinus)

In the process of focused coding, we develop a codebook. For each analytical theme, the codebook contains a definition, guidelines for when to code and not to code a thread as an instance of the theme, and a leading example of a thread that instantiates the theme. The codebook serves two purposes in the interpretation and grounding phase and for the integrated analysis as a whole. First, the codebook explicates and summarizes the links between theoretical concepts and empirical observations that we establish through our qualitatively analysis. As such, the codebook increases the transparency of our qualitative analysis and serves as a means to assess its validity (MacQueen et al., 1998; Frederiksen et al., 2023, pp. 161–163). Second, we use the definitions and guidelines in the codebook for closed coding of our test set of 60 threads. In the process of closed coding, we apply the themes developed through focused coding to annotate the test set without engaging in in-depth analysis. Importantly, the annotated test set was held out of the qualitative analysis itself, thus making it useful for the purpose of validating the dictionary-based classifier that we implement in the third phase of our analysis design (Carlsen & Ralund, 2022).

3.2.3 Classification and Measurement

The purpose of the third phase in our analysis design is to test whether the analytical themes discovered and developed in the first and second phase generalize to the entire corpus. To achieve this, we first use the dictionary to classify all threads in the corpus into analytical themes. To assess accuracy, we use the manually annotated test set to validate the dictionary. This classification procedure allows us to investigate how frequently extremist rhetoric in its different forms occurs in the *incels.is* forum. A related and theoretically relevant question is how much attention and interactivity elements of extremist rhetoric generate in the forum. To get an insight into this aspect of our research question, we use the predicted thread labels as inputs in a linear regression.

3.2.3.1 Dictionary Method (Sarah)

A dictionary method labels texts based on a human-generated set of rules. The advantage of using a dictionary method over more sophisticated supervised learning methods is that it does not require manual labeling of a training set. This reduces the amount of labor required for classification of a large corpus while retaining its grounding in human annotation. Further, with domain-specific knowledge of language use, dictionary methods can deliver good performance

in classification tasks. Because we achieve in-depth knowledge of language usage in incel culture through qualitative analysis, we believe a dictionary method to be a well-chosen classifier in the context of our analysis design (Grimmer et al., 2022, pp. 178–180). To implement the dictionary-based classifier, we assign a weight to each entry in the dictionary. The weights reflect the importance of words in the dictionary to their associated analytical themes. Using the dictionary entries and their weights, we compute a set of theme-specific scores for all threads in the corpus. We then classify each thread into the analytical theme for which it has the highest score, subject to a lower threshold. Threads that have scores below this threshold for all analytical themes are classified into a thematic residual category. To validate this dictionary-based classification method, we compare predicted to hand-coded labels in the manually annotated test set by computing an error rate (Carlsen & Ralund, 2022; Grimmer et al., 2022, pp. 180–183). Finally, we use the predicted thread labels to display the relative frequencies of the analytical themes in the entire corpus in a table. This allows us to measure the general prevalence of different forms of extremist rhetoric in the *incels.is* forum.

3.2.3.2 Linear Regression (Sarah) To investigate the amount of attention and interactivity different forms of extremist rhetoric generate in the *incels.is* forum, we estimate two linear regressions of the form

$$Y_i = \beta_0 + \sum_{j=1}^{J-1} \beta_j X_{ij}$$

where X_{ij} are a set of dummy variables equal to one if thread i is classified into analytical theme j. The excluded reference category is the thematic residual category. In the first version of this regression, Y_i denotes the number of views of thread i. In the second version, Y_i denotes the number of comments in thread i. We use the number of views of a thread to operationalize the amount of attention it generates and the number of comments in a thread to operationalize the amount of interactivity it generates. Thus, the coefficients β_j tell us how much attention and interactivity each form of extremist rhetoric generate in the *incels.is* forum relative to the thematic residual category. Importantly, the estimates for these coefficients have a specific role in the interpretation of the integrated analysis as whole. We briefly discuss these principles of integrated interpretation in the context of our research question and analysis design next.

3.2.4 Integrated Interpretation (Sarah)

An important aspect of the theoretical account presented in this assignment is the notion that the components of extremist ideologies can be separated into core concepts, which are necessary and fundamental building blocks, and peripheral concepts, which may or may not be articulated in extremist worldviews. In the context of our study of incel culture, the theory thus generates expectations pertaining to, first, the content of the discourses in the *incels.is* forum and, second, the centrality of these different discourses for social interaction in the forum. Our analysis is designed to confront theoretical expectations in these different areas. Whereas the interpretation and grounding phase addresses the content of the discourses, the classification and measurement phase addresses the centrality of these discourses for social interaction. Accordingly, the process of moving from empirical results to conclusions relies on a theory-driven form of integrated interpretation where the different phases in our analysis design play distinct and complementary roles in answering our research question (Frederiksen et al., 2023, pp. 151–154). Moreover, the twofold correspondence between theory and analysis design in our study has the implication that we are actively facilitating discovery of surprising empirical patterns pertaining to at least two aspects of extremist ideology, namely the contents of different forms of extremist rhetoric and their hierarchical organization as ideological components (Brandt & Timmermans, 2021; Karell & Freedman, 2019; Timmermans & Tavory, 2022, pp. 1–4).

4 Results

4.1 Immersion and Discovery (Noel)

Through the process of netnographic immersion, we discovered that the theoretical core concepts of extremist ideology can reasonably be used as analytical themes for studying incel culture. While discussions in the *incels.is* forums relate to a broad range of topics, the core concepts of extremist ideology appear frequently and across otherwise diverse contexts. We therefore decided to work with othering, victimhood, and monopoly on truth as preliminary analytical categories. The immersive data operations also generated preliminary seed word lists consisting of primarily self-invented and idiosyncratic language pertaining to each of these categories. For example, we discovered that incels use the word "foid" as a derogatory term for women, and mention "genetics" often

when contemplating their victimhood.

The output of the LDA model is displayed in appendix A along with our chosen label for each of the 20 topics. The LDA model confirmed our initial notion that the core concepts of extremist ideology appear within our corpus. First, the model identified topics that we annotated as "Chad" traits, prostitution, race, social hierarchy, slurs, and gender, that we group together under the analytical theme of othering. Second, the model identified topics that we annotated as physical appearance, virginity, suicide, and oppression that we group together under the analytical theme of victimhood. Finally, the model identified topics that we annotated as collectively held beliefs and nature and society that we group together under the analytical theme of monopoly on truth. Furthermore, we used the output of the LDA model to expand our lists of seed words. For example, we expanded our seed word list for the theme of victimhood with the word "bully" which appears in the LDA topic that we labeled as oppression.

Using Word2Vec, we expanded our lists of seed words to a set of keyword lists. Through a process of trial and error, we decided to perform the expansion operation by including the 10 most similar word of each seed word based on cosine similarity in the lists of keywords. The process left us with a set of keyword lists shown in appendix B.

4.2 Interpretation and Grounding

4.2.1 Othering (Noel)

Through our close reading analysis, we mapped variation relating to the concept of othering. Othering manifests in discussions of minority groups as well as in the delineation of an in-group and an out-group based on a looks-based social hierarchy. We also observe the concept of purity where users in the forum contend qualifiers of being a true incel and campaign for the site to stay uncorrupted by “normies” or non-incel individuals.

4.2.1.1 In-Group and Out-Group (Noel) Using the index case, we provide an example of the type of language used between incels to highlight their designated out-group. In the thread titled "Ultimate Cuck", a user relays their experience of being rejected by a college woman:

> *As all her Stacy friends and other normies gave me these looks of disgust and mockery, I noticed her talking to some Chad. And laughing. I went to*

the bar to get some beers, and when I came back there she was, kissing him. In a matter of minutes, she had deemed this Chad worthy of trading saliva with her.

It is common for incels to use looks-focused language to establish an out-group in alignment with preconceived notions of social hierarchies. In this hierarchy, the most attractive individuals, "Chads" and "Stacys", reside at the top due to their ability to obtain a romantic partner. Next are "normies" who are, in the eyes of incels, average to society via looks and personality. Finally, incels place themselves at the bottom because they qualify themselves as ugly and unsociable, thus unable to achieve a romantic partnership.

While there exists a social hierarchy based on looks and social skills, incels still accentuate that women are lesser than men no matter the social status or attractiveness throughout their threads. "Stacys" are perceived as higher in status because of their ability to attract a partner. However, dehumanizing words such as "foid", "landwhale", and "meat toilet" are used in-place of "woman". A thread titled "Foids deserve to get stalked" provides an example:

I guess this is an unpopular opinion especially me being an incel, but anyway, I used to feel bad for Foids and get creeped out and disgusted at stalkers for stalking Foids, however, now I think they deserve to get stalked on tbh.

The author's use of the word "foid", or female android, is part of a dehumanizing practice in which women are portrayed as heartless robots who prevent incels from achieving their goal of obtaining sexual and/or romantic relationships. As incels see this goal as unattainable, a blame is placed on women and thus an us vs. them narrative is established. This blame can also justify violent and violating actions as highlighted by the author of the post. In the case of using othering-type language for racial minorities, we do not include that as a separate sub-theme due to the lack of differentiation of race between in-group and out-group.

4.2.1.2 Purity (Noel) Othering language is sometimes collectively directed inwards in the incel community. In these cases, incels display a preoccupation with purity in alignment with expectations generated by theories of extremism. Discussions of what constitutes a real incel take place in the *incels.is* forum, and commonly, accusations of being a fake incel are exchanged. Words such as "normies", "LARPer",

and "fakecel" are used to distinguish non-incel intruders whose attendance within the site causes mass distress. In the post "Defining Fakecelism", the author discusses the characteristics of a fake incel:

> *So, "fakecel" is a term that's flung around like shit on a monkey's cage[...]There's enough areas of overlap between a fakecel and a LARPer, a fakecel and a volcel, or a fakecel and a mentalcel to, I think, warrant some community discussion. So, how would you define a fakecel?*

Incels are distinguishing an in-group and out-group within their own community based on varying descriptions of what a true incel is. As shown in the example there is confusion to the community as to who constitutes a "fakecel" or "LARPer", lending to a collective crisis of normie infiltration. Paired with the strong forum rules, the preoccupation with the purity of the site and their efforts to distinguish themselves from the "normies" is a clear focus of the incels.

4.2.2 Victimhood

Besides the othering practices described above, we observe expressions of victimhood in the forum. The expressions vary in their emphasis on the main cause of grievances. We discovered two often appearing causes: an internal cause, being genetics, and an external cause, being the status, success or privilege of the out-group, mostly women. In addition, in both of these forms of victimhood, we observed a tendency to outline comprehensive analyses of society with underpinning notions of conspiracy and catastrophe. Lastly, we observed a theme related to but separate from victimhood, being self-hatred.

4.2.2.1 Victims of the Out-Group's Success (Signe) Through the index case, we aim to illustrate a typical example of how victimhood is represented in incel culture. In the thread, the author argues that gender equality by law would be disadvantageous for men as women are the naturally privileged gender:

> *If both genders were equal by law, there would be a massive power gap between men and women[...]Women are the choosers of sex[...]Foids would all share the same elite chads[...]Incels shouldn't strive for gender equality since biology just doesn't allow for it.*

The example reveals a sense of injustice, suggesting that women's natural privileges allows them to unjustly distribute sex only to "elite chads". This aligns with the perception in the community which claims that female sexuality inherently strives for men with the highest attractiveness ratings. The quote thus demonstrates incels as victims of women's sexuality and power, linking incels' victimhood to the privileges of the out-group, women. Moreover, the quote implies that incels are entitled to sex and that women have the power to distribute it among men. While feeling deprived of intimate relationships is legitimate, framing it as injustice echoes the extremist structure in its exaggeration of female power and its dehumanizing understanding of women's rights to their own body.

Besides women's easy access to sexual relationships, the index quote depicts a widely spread view of how women navigate through life with few obstacles. This is also seen in a thread where the author asks if people would choose to be born as a woman or a man. Several posters reply by highlighting the advantages and privileges that women possess and the derived societal injustice:

> *I'd rather be a woman. They live life on negative difficulty. There are some disadvantages but it doesn't matter. Why? Because literally every flaw you have as a woman just gives you more points[...]meanwhile every flaw you have as a man gives you like -50, fuck being a man. Just fucking lol when these sluts cry about anything, they have it so good tbh.*

The author perceives a general cultural and societal imbalance: women are rewarded for their flaws while men are punished. Incel's low societal esteem is repeatedly contrasted with women's favorable social conditions, reflecting a strong sense of injustice related to the out-group. This is also why they ridicule the idea of structural gendered oppression as shown in the latter part of the quote above and in this comment: *"Things are only taken seriously when it affects toilets. Since toilets cannot be incels, society will always ignore our issues."* Using "toilets" as a derogatory slang for women, the user emphasizes how the salience of women's issues are attended to in an unjust and disproportional manner, at the expense of incels' issues.

4.2.2.2 Victims of Genetics (Signe) Another variation of victimhood regards being a victim of genetics. We observed a hyperfocus on the appearance determined by genetics, such as facial features, bone structure, and body physique. This includes

an emphasis on genetics as the cause of one's lack of romantic relationships and social status as seen in the following example:

> *Due to the rules of biology, young women have a very high baseline SMV [sexual market value] that allows them to easily get sex and relationships regardless of looks[...]For a woman to be attracted to a man, he needs to have a distinguishing quality, and that's why appearance is so essential as a man.*

Here, it is described how the impact of genetics is skewed between men and women which also relates to a sense of injustice. This aligns with our discussion in the theory section that in extremist beliefs, the in- and out-groups are often situated in a struggle determined by greater forces, such as nature, and in this case, in the form of biology and genetics. The perceived importance of genetics in one's success in romance is also prevalent from this quote:

> *I was always an ugly child. I had protruding front teeth, bug eyes, I was tall for my age but so thin that I looked anorexic[. . .]In spite of this [I was still hoping] that one day I might have a girlfriend[. . .]38 years later I sit here every night alone in my little flat, a string of rejections and humiliations behind me.*

We see how the user attributes his lack of a girlfriend and lifelong rejection to his perceived "ugliness," proving wrong those that had claimed that women would value his personality and sense of humor over looks.

4.2.2.3 Conspiracy and Catastrophe(Sarah) Another form of victimhood is manifested in the form of creating imagined narratives, often conspiratorial, that the in-group perceives as reality. In the following quote, the user portrays a pessimistic view that for an incel, all non-incel relations are false friendships:

> *The average person you see on the street, at work, in college or anywhere else wants you to rope ASAP[...]Oh you have a Chad friend? He's not your friend, you're the male equivalent of a fat friend of his[...]You talk to normie men at work who treat you fine? Well they're being polite and want you to kill yourself.*

This user is conspiring that all members of the out-group want the incels eradicated and want all incels to commit suicide. This post assumes victimhood in the incel community from all individuals of society, indicating that in every social situation the incel engages in, they are victimized and secretly wanted dead. The user above is creating a narrative with minimal basis and projecting it onto the whole of the out-group and on every interaction they have in the future.

In addition to users sharing conspiracy, the incel community often uses a catastrophic view of the future to prove their oppression from the out-group. In these narratives, the out-group, or oppressor, is society and the rules that it enacts upon the incel community:

> *In the trans humanism future, we the ugly males or the "incels" would be caught and brutally attacked and eradicated or be made slaves, meanwhile ugly females won't be eradicated[...]and the Chads will engage in polygamy impregnating more and more Stacey's and begetting beautiful childrens on them.*

These imagined catastrophes can be used to justify present violence by the incel community to protect themselves from this supposed oppression they may be subjected to in the future. The presence of both conspiracy, reflecting incel belief on the current reality, and catastrophe, reflecting theories about the future, both exist within the corpus to exacerbate the victimhood the incels place upon themselves.

4.2.2.4 Self-hatred (Sarah) The variations of victimhood described so far align with our theoretical understanding of extremism. In all these variations, victimhood is derived from the social hierarchies as the underpinning cause. However, we also observed a different representation of the incels' miserable situation, namely in the form of self-hatred. This self-hatred is present in how incels engage in self-deprecating language. While the emotion of self-hatred resembles the feeling of victimhood in its misery, self-hatred differs in being purer, being more inward oriented, and more self-imposed. For self-hatred, the emphasis is less on how social dynamics are pushing incels into unfavorable victim positions and more on their own self-perception:

> *At my age (18) and being born in a shitty country worse than some in Africa, I can say that my life is not going to get better, but to get worse. I am even more subhuman than the subhuman shitskins of my country and to make*

matters worse I have OCD and diagnosed depression. This is the first time I join a forum, my IQ is so low that I don't know how to write a text clearly.

This incel shows a strong antipathy towards his circumstance, outer appearance, and intellectual capabilities, even referring to himself as "subhuman" - community slang for an ugly person. They view their situation as so dire that they feel hopeless about their future - a hopelessness seen commonly throughout the forum. In the following quote, the user's self-hatred leads him to believe he is fundamentally inferior and unworthy of a partner due to his perception of himself as a "gigasubhuman", an exaggeration of the more commonly used "subhuman":

I hate women for being parasites but not for rejecting me. I just feel sad that they don't. I wouldn't date myself either because I'm gigasubhuman. I'm a low T, sexually untouchable, diseased shitskin subhuman who is incapable of achieving anything in life. I am the worst of the worst. I don't deserve to mate and breed. If a woman took pity and bore my offspring, it would destroy her life.

In this case, his victimhood is not derived from previous rejections or his treatment by women in the past, but a self-deprecation he is placing upon himself. While some incels ascribe their miserable position to external forces and display a hatred of the oppressor, we have seen that many others feel as though their inceldom begins internally and is a plight they are born into. Many incels hate the fact that they are incels and wish they could break out of their fate of celibacy.

4.2.3 Monopoly on Truth (Stinus)

Here, we map rhetorical patterns in the forum revolving around true and false suppositions about human nature and fundamental principles of social order. In doing so, we show that the belief systems of incels align with tendencies in extremist ideology to claim a monopoly on truth.

The concepts of the blackpill, redpill, and bluepill are central to collectively held beliefs about human nature and social order in the incel community. Etymologically, these concepts originate from the 1999 science-fiction movie The Matrix where the protagonist Neo is offered a blue pill that lets him stay in a state of fake consciousness or a red pill that lets him wake to a dystopian reality where humanity is enslaved by

machines. In the *incels.is* forum the blackpill - an expanded version of the redpill - refers to the belief that life trajectories are genetically determined. This belief most commonly surfaces in discussions about romantic relationships as this is the focal issue around which incel identity is constructed:

> *No, blackpilled means an understanding that how good looking you are is what will ultimately define your romantic success, no matter how charismatic or sociable you are, particularly as a man.*

As argued throughout our qualitative analysis, genetics and looks play central roles in most aspects of incel ideology. Hence, the concept of the blackpill does not only relate to men's dating success but serves as a fundamental structuring principle of the social world that explains social hierarchies, the victimization of incels, and their tendency to hate themselves. The blackpill is collectively accepted as an uncontestable and comprehensive truth in the incel community:

> [Poster]: *You know the worst thing about this world? People are afraid to see the truth.*
> [Commenter]: *100%. Blackpill doesn't do much more than observe and point out the nature of humanity.*

The blackpill is contrasted to the bluepill. The bluepill is the naïve rejection of genetic determinism. Being bluepilled is a characteristic of people who do not have "*[...]thick enough skin to hear the truth* [and believe in a] 'just world' fallacy[...]". The juxtaposition of the blackpill and the bluepill consolidates the postulated objectivity of the blackpill. As such, it constitutes a discursive claim to a monopoly on truth made collectively by the incel community. This rhetorical pattern surfaces in a thread about the determinants of personality. The poster claims that personality is shaped by appearance through society's responses to individuals with certain physical traits. A commenter agrees to these propositions but contemplates that most people would not understand the underlying logic. Another commenter then argues that people outside the incel community live in a state of false consciousness and therefore cannot comprehend the true nature of humanity and society:

> [Poster]: *If I were to explain this [development of a hateful personality] to generally anyone they would say I did it to myself.*
> [Commenter]: *They're religiously bluepilled. It'd be like an atheist trying to convince a Christian there is no God.*

4.3 Classification and Measurement (Sarah)

Through focused coding, we developed a dictionary mapping subthemes to words, shown in table 1. We associated dictionary words with weights reflecting how often they are used in each subtheme. Using the weighted terms from the dictionary, we then classified forum threads into subthemes as described in section 3.2.3.1. We classified threads that received a score for all subthemes below the value of 1 into a thematic residual category. In order to validate the dictionary-based classifier, we built a codebook and manually coded a test set of sixty forum posts using the codebook as a framework. The codebook is displayed in appendix C. The human-coded entries were then compared to our computed classifier, resulting in an error rate of 56.7%.

Table 1: Dictionary

Subtheme	**Othering**	**Victimhood**	**Truth**	**Self-hatred**	**Purity**
Words	1 * foid 1 * chad 1 * stacy 0.5 * chadlite 0.5 * femoid	0.5 * ugly 0.5 * cope 0.75 * privileged 0.5 * standards 1 * feminized 1 * ignore 1 * oppressed	1 * blackpill 1 * bluepill 0.5 * redpill 0.5 * antinatalism	1 * subhuman 1 * hate 1 * suicide 0.5 * flawed	1 * truecel 0.5 * normie 0.75 * larping 0.75 * trolling 1 * infiltrator 1 * fakecel

Table 2: Relative Frequencies of Subthemes

Subtheme	**Relative Frequency (%)**
Othering	18.20
Victimhood	2.84
Monopoly on Truth	4.00
Self-hatred	5.90
Purity	1.08
Residuals	67.98

Table 2 displays relative frequencies of the subthemes in the full corpus as labeled by our dictionary-based classifier. The results support the conclusion that all the subthemes are relevant analytical categories from a corpus-wide perspective. In combination, the concepts of othering, victimhood, monopoly on truth, and purity are featured in 25%

of the threads in the *incels.is* forum. The wide-spread presence of these subthemes is indicative of extremist tenets within the forum as a whole. Othering is the most frequently occurring of the themes within the corpus. While purity and victimhood are still present, they occur on a less frequent basis. The concept of self-hatred, not a component of extremist ideology, is the second most frequently occurring of the five subthemes within the *incels.is* forum. This suggests that self-hatred is a salient tenet of the incel worldview.

Table 3: Linear Regression Results

	Views	Comments
	(1)	(2)
Othering	150.3***	0.7***
	(6.6)	(0.1)
Purity	253.6***	5.4***
	(24.2)	(0.4)
Self-hatred	156.5***	1.7***
	(10.7)	(0.2)
Truth	130.2***	0.5**
	(130)	(0.2)
Victimhood	284.7***	2.1***
	(15.3)	(0.2)
Intercept	889.8***	29.8***
	(3.0)	(0.0)
Observations	223760	223760

*Note: Standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01*

Table 3 shows the results of running the linear regressions on the corpus as specified in section 3.2.3.2. The results indicate that when threads feature one of the subthemes, they are likely to generate increased attention and interactivity through comments and views as compared to the residual category. The high prevalence of all the subthemes and their tendency to generate engagement in the forum suggests that categorizing the

incel worldview writ large as a case of extremist ideology is justified. While purity and victimhood appear less frequently in the corpus than the remaining subthemes, they generate higher engagement. In alignment with theories of extremism, this suggests that the feeling of victimhood and the preservation of the in-group identity are central elements in incel culture. On the other hand, while othering is the most frequently featured subtheme in the corpus, it generates the least amount of engagement. This is unexpected as othering is a core concept in the structure of extremist ideology. Discussions about self-hatred generate the third most engagement among the subthemes. The observation that discussions about self-hatred generate a substantial amount of attention and interactivity is surprising in the sense that it is an unexpected component in an extremist worldview.

5 Discussion and Conclusion (Stinus)

Our digital mixed methods analysis supports the conclusion that a range of tenets in incel culture can reasonably be categorized as extremist. We mean this in the sense that rhetorical patterns displayed in the incel community instantiate the theoretical components of extremist ideology described in the literature. First, we find that incels believe in rigid social hierarchies and tend to other individuals that are not part of the incel community using dehumanizing language. This stream of discourse is associated with a preoccupation with purity. A recurring theme in the forum is thus whether its members are true incels or imposters. Second, we find that incels discuss feelings of victimhood extensively and attribute their victimization to a set of social and biological systems that oppress members of the incel community while privileging other individuals. Furthermore, feelings of victimhood are often grounded in beliefs about conspiracies among non-incels and connected to projections of impending catastrophe. Finally, incels present their views on human nature and social order as the ground truth and discursively construct opponents of their arguments as suffering from a state of false consciousness. Our quantitative results suggest that, not only are these themes present throughout the *incels.is* forum, they also generate more engagement in terms of viewership and written comments than other content. These findings contribute to debates in the literature about whether to conceptualize the worldview of incels as extremist (O'Hanlon et al., 2024).

While most of our findings make sense in the light of dominant theories of the

structure of extremist ideology, we discovered a stream of discourse that our theoretical priors cannot fully account for. A widely discussed topic in the *incels.is* forum is self-hatred. Importantly, self-hatred can be meaningfully thought of as a central part of the collective identity of incels as it is a commonly debated topic in the forum with apparent logical relations to other tenets in incel culture. However, self-hatred does not readily confine to any of the core or peripheral concepts of extremist ideology. Because our remaining results conform to theories of extremism, the finding that self-hatred is a prevalently discussed topic among incels is anomalous in the light of these theories. It therefore presents us with an opportunity to confront the second part of our research question by drawing an abductive inference. Specifically, we propose the following hypothesis on probation: *Self-hatred is a peripheral concept in the structure of extremist ideology.* In other words, self-hatred is a contingent component of extremist ideology that does not necessarily configure in its structure but instead prescribes specific meanings to the core concepts and contributes to separate forms of extremism when present. If this hypothesis were true, the full range of empirical findings from our analysis would make sense through the lens of theories of extremism. In turn, augmenting these theories by including self-hatred as a peripheral concept would lend them credibility by allowing them to cover the universe of cases of extremism, including incel culture (Karell & Freedman, 2019; Timmermans & Tavory, 2012; Brandt & Timmermans, 2021).

Following the abductive research cycle, our hypothesis on probation should jump-start further research using both inductive and deductive modes of inquiry. Working inductively, we would return to our data to look for patterns that speak to the abductive inference. Working deductively, we would derive specific hypotheses about related cases to proceed with collection of new data (Timmermans & Tavory, 2022, pp. 155–157). While we leave these empirical efforts to future research, past work on extremism does in fact provide some corroboration to our hypothesis on probation. In the wake of the 9/11 attacks, a strand of psychological research emerged that investigated the traits and beliefs of Islamist suicide terrorists who pursue martyrdom. A recurrent finding in this literature was that suicide terrorists are driven by ingroup-outgroup enmity, victimhood narratives, and an extreme willingness to sacrifice themselves for what they believe to be the one true cause (Post et al., 2009). These observations lead to the conclusion that suicide terrorists can be reasonably considered prototypical extremists according to the dominant theories reviewed in this assignment. However, based on an extensive investigation of primary sources such as suicide notes, manifestos, and diaries, Lankford (2014) argues that an additional characteristic of suicide terrorists is

that they display conventional suicide risk factors such as mental illness, psychological pain, and personal crises. For example, an analysis of a manifesto written by the 9/11 ringleader Mohammad Atta reveals that he was suffering from depression and social isolation (Lankford, 2012). Bobadilla (2014) goes one step further to argue that the stereotypical psychological profile of a suicide terrorist is the vulnerable narcissist. As vulnerable narcissists, suicide terrorists often display a combination of inflated and unstable self-esteem, aggressive tendencies, and extreme social anxiety. In the words of Bobadilla (2014, p. 365): *"Self-destructive killing requires the desire and capability for suicide and murder: in short, a unique mixture of self-loathing and antagonism"*. Thus, self-hatred is an important component in the belief systems of suicide terrorists that lead them to engage in self-destructive extremist violence. Hence, this strand of research on the psychology of suicide terrorists resonates with our findings that a central element in the collective identity of incels is self-hatred. By heuristically lumping together Islamist suicide terrorist and incels, we can support our argument that self-hatred is a common denominator in some forms of extremism (Abbott, 2004, pp. 144–146). This corroborates our hypothesis on probation and suggests that future research should seek out related cases to map out the conditions and consequences of self-hatred as a trait of violent extremists.

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A LDA Output

Topic Number	**Output**	**Chosen Label**
1	'social', 'also', 'may', 'less', 'people', 'physical', 'study', 'likely', 'one', 'health', 'often', 'however', 'attractiveness', 'mental', 'sexual', 'found', 'self', 'many', 'due', 'research'	Chad traits
2	'uwsl', 'children', 'kids', 'mother', 'child', 'wife', 'married', 'family', 'father', 'parents', 'marriage', 'husband', 'son', 'baby', 'marry', 'daughter', 'single', 'man', 'divorce', 'onlyfans'	Family life
3	'work', 'job', 'iq', 'working', 'games', 'jobs', 'college', 'high', 'play', 'low', 'neet', 'playing', 'parents', 'hours', 'game', 'sports', 'studying', 'money', 'career', 'degree'	Strategies to escape inceldom
4	'body', 'face', 'hair', 'eyes', 'look', 'fat', 'skin', 'eye', 'weight', 'facial', 'gym', 'wear', 'faces', 'features', 'nose', 'skinny', 'wearing', 'clothes', 'teeth', 'lbs'	Facial appearance
5	'sex', 'love', 'sexual', 'porn', 'dog', 'pleasure', 'relationship', 'desire', 'virgin', 'dogs', 'virginity', 'romantic', 'vagina', 'sperm', 'ai', 'body', 'experience', 'sexually', 'affection', 'woman'	Virginity

Continued on next page

Topic Number	**Output**	**Chosen Label**
6	'nt', 'autistic', 'suicide', 'autism', 'er', 'voice', 'anime', 'curious', 'cel', 'chin', 'tranny', 'fuel', 'jesus', 'trannies', 'autist', 'commit', 'sentence', 'saint', 'christ', 'church'	Suicide
7	'https', 'com', 'view', 'www', 'reddit', 'comments', 'youtube', 'watch', 'share', 'utm_source', 'utm_medium', 'youtu', 'tiktok', 'video', 'utm_name', 'android_app', 'inceltear', 'twitter', 'context', 'utm_term'	Social media
8	'height', 'short', 'tall', 'manlet', 'average', 'taller', 'small', 'size', 'face', 'dick', 'cm', 'jaw', 'mogged', 'ft', 'frame', 'inches', 'shorter', 'penis', 'balding', 'big'	Bodily appearance
9	'got', 'said', 'one', 'school', 'back', 'time', 'friends', 'girl', 'years', 'told', 'first', 'went', 'day', 'started', 'saw', 'friend', 'last', 'made', 'wanted', 'ago'	School
10	'money', 'pay', 'food', 'eat', 'buy', 'surgery', 'water', 'free', 'eating', 'car', 'drugs', 'drink', 'paying', 'escorts', 'homeless', 'escort', 'paid', 'rich', 'expensive', 'dollars'	Prostitution
11	'like', 'even', 'would', 'get', 'people', 'life', 'want', 'think', 'know', 'never', 'one', 'make', 'see', 'go', 'much', 'way', 'really', 'time', 'feel', 'good'	Hopes and ambitions

Continued on next page

Topic Number	Output	Chosen Label
12	'old', 'year', 'age', 'click', 'expand', 'video', 'young', 'years', 'able', 'display', 'older', 'browser', 'younger', 'spoiler', 'movie', 'teen', 'adult', 'teenage', 'teens', 'aged'	Porn
13	'toilets', 'rape', 'toilet', 'raped', 'bullying', 'bully', 'dark', 'bullied', 'bullies', 'boyos', 'vm', 'beaten', 'buddy', 'nofap', 'minecraft', 'triad', 'virtue', 'filthy', 'gang', 'roast'	Oppression
14	'white', 'black', 'asian', 'race', 'ethnic', 'pill', 'curry', 'american', 'blue', 'country', 'indian', 'whites', 'countries', 'african', 'chinese', 'china', 'curries', 'brown', 'rice', 'races'	Race
15	'foids', 'chad', 'ugly', 'foid', 'guys', 'get', 'guy', 'looks', 'looking', 'girls', 'chads', 'even', 'good', 'average', 'normie', 'look', 'dating', 'girl', 'high', 'personality'	Social hierarchy
16	'fucking', 'fuck', 'shit', 'hate', 'ass', 'pussy', 'whores', 'bitch', 'whore', 'stupid', 'dick', 'disgusting', 'faggot', 'retarded', 'dumb', 'gay', 'fucked', 'bitches', 'faggots', 'slut'	Slurs
17	'women', 'men', 'woman', 'man', 'female', 'male', 'females', 'males', 'gender', 'sexual', 'feminist', 'attractive', 'dating', 'feminism', 'sexually', 'feminists', 'attracted', 'many', 'young', 'masculinity'	Gender

Continued on next page

Topic Number	**Output**	**Chosen Label**
18	'kill', 'police', 'co', 'news', 'violence', 'killed', 'uk', 'august', 'prison', 'murder', 'court', 'crime', 'elliot', 'according', 'shooting', 'death', 'arrested', 'wikipedia', 'said', 'law'	Mass shootings
19	'incels', 'incel', 'post', 'forum', 'blackpill', 'online', 'thread', 'community', 'media', 'site', 'internet', 'posts', 'users', 'videos', 'account', 'threads', 'banned', 'read', 'content', 'posted'	Collectively held beliefs
20	'world', 'society', 'us', 'human', 'nature', 'power', 'humans', 'evil', 'god', 'born', 'must', 'modern', 'control', 'become', 'system', 'state', 'based', 'war', 'genes', 'people'	Nature and Society

B Seed Words, Keywords, and Dictionary

Analytical Theme	**Seed Words**	**Keywords**
Othering	foid, chad, normie, fakecel, truecel, larp	'foid', 'femoid', 'foids', 'female', 'woman', 'roastie', 'girl', 'femoids', 'chick', 'chad', 'chads', 'gigachad', 'chadlite', 'incelchad', 'tyrone', 'stacy', 'latrel', 'overstacey', 'slops', 'normie', 'normalfag', 'normies', 'norman', 'nomrie', 'normiesoy', 'normiewølf', 'normiediocel', 'fakecel', 'fakecels', 'volcel', 'larper', 'mentalcel', 'truecel', 'noalcohol', 'rper', 'infiltrator', 'truecels', 'incel', 'khhv', 'trucels', 'tkhhhv', 'traittruecel', 'trueceli', 'larp', 'larping', 'troll', 'larps', 'larpcopexodius', 'trolling', 'brigader'
Monopoly on Truth	blackpill, bluepill, redpill, truth, delusional	'blackpill', 'blackpills', 'bluepill', 'pill', 'blackpillthey', 'blackpilled', 'racepill', 'blackpilli', 'pillpill', 'bluepilled', 'redpill', 'bluepills', 'redpilll', 'bluebar', 'prototrannies', 'wholheatedly', 'redpill', 'pua', 'mgtow', 'redpilled', 'redpillers', 'trp', 'repill', 'muuuuuuuh', 'truth', 'truths', 'unvarnished', 'facts', 'reality', 'truthi', 'truthyes', 'factsbecause', 'waifupill', 'delusional', 'deluded', 'cutehunter', 'retarded', 'incelscopetv', 'articlejfl', 'thintard', 'delusion', 'delutional'

Continued on next page

Analytical Theme	**Seed Words**	**Keywords**
Victimhood	genetics, bully, virgin, autism, gynocen-trism, humiliate, smv, system	'genetics', 'genes', 'genetic', 'enviourment', 'geneticstldr', 'genetically', 'looksyour', 'themnonsense', 'englandgenetics', 'bully', 'bullies', 'bullying', 'bulllies', 'mock', 'bullied', 'guessingzombie', 'insulty', 'bullyasiacel', 'losersguys', 'virgin', 'virgins', 'kissless', 'handholdess', 'khhv', 'danceless', 'handheldless', 'lookless', 'hugless', 'khhvm', 'autism', 'asperger', 'aspergers', 'autistic', 'adhd', 'diagnosed', 'hfa', 'auitism', 'schizophrenia', 'symptomology', 'gynocentrism', 'hypergamy', 'feminism', 'eastoid', 'disposibility', 'headquaters', 'lovedmainländer', 'manocentrism', 'widly', 'gynocentric', 'humiliate', 'insulte', 'mock', 'suzuka', 'bullly', 'mandatorly', 'himvisit', 'proceeed', 'embarrass', 'botit', 'smv', 'smvslavcel', 'attractivess', 'smvjust', 'svm', 'smvif', 'smva', 'smvi', 'artifically', 'smvit', 'system', 'systems', 'systemi', 'indiaincline', 'obsolte', 'strenghten', 'bullyimg', 'necessariting'

C Codebook

C.1 Othering

Code: 1
Brief Definition: Creating a narrtive of "us vs. them" or in other words an in-group and out-group
Full Definition: Othering refers to the creation of an in-group and an out-group or an "us" vs. "them" narrative. The in-group share common beliefs, traits, and practices where the out-group is looked down upon for not holding truth to such ideas. Normally the out-group is refered in a demeaning and dehumanizing way that emphasizes that that there are differences between groups.
When to Use: Use this code when the writer insinuates a difference between themselves/incel group and another group of individuals. This can be achieved through blantant expression of the existence of an outgroup and/or using special (mostly made up) language inplace of identifiers.
When to Not to Use: Do not use this code if the the post talks about containing purity within the site or in within the incel communnity meaning if the users are "true incels". Do not use if the writer is differentiating different world views so that ideas of truth are discussed. Do not use this if the writer only discusses self-hatred.
Example: *[..]after riding miles of Chadlite and Chad cock. She's had over 10 boyfriends that I know of (starting at the age of 11-12, all of them were tall), probably fucked 50+ men. At 25 or so she suddenly turned religious and found herself a "religious" 6'8" Chadlite*

C.2 Victimhood

Code: 2
Brief Definition: The feeling of victimhood
Full Definition: Relates to the feeling of victimhood caused by the social structures of society alongside with the feeling of being oppressed. The victimhood is often framed as being caused by the success of an out-group, often women or 'good-looking' men or "chads", or by one's genetics. It can also involve larger, often dystopian, analyses of the state of society and how this stands in the way of possible success of the in-group.
When to Use: Apply this code when it describes the victimhood state that incels are in, either due to the status of other groups or one's own genetic disposition.

When to Not to Use: Do not apply this code to a post if it only contains hateful descriptions of an out-group without any (indirect as direct) reference to how this is causing the miserable life situation of the incels. While victimhood is closely tied to the perceived unfair dynamics between the in- and the out-group and to the fact that the out-group is deemed intrinsicially harmful, posts that only demonize an out-group should not be coded as "Victimhood" (see OTHER).
Example: *Things are only taken seriously when it affects toilets. Since toilets cannot be incel, society will always ignore our issues.*

C.3 Monopoly on Truth

Code: 3
Brief Definition: Two opposing worldviews of genetic determinism vs. agency
Full Definition: Contrasting belief in genetic determinism with belief in agency in romantic relationships and life in general. The former is referred to as the 'blackpill' while the latter is referred to as 'the bluepill'. 'The redpill' refers to a belief system closely related to the blackpill. However, it is less drastically deterministic. The term 'blackpill' is often used to frame the experiences of incels as grounded in objective truth about human biology. Further, the realization of 'the blackpill' is sometimes portrayed as a burden that causes depression. Outsiders who have not yet discovered that genetic determinism in a fundamental principle of social life are referred to as 'bluepilled'.
When to Use: Apply this code when a user defines 'the blackpill' or 'the redpill', i.e., when a user designates a concept related to genetic determinism as a structuring principle of the social world. Apply this code when a user argues against some proposition by refering to 'the bluepill'. Also apply this code if a user argues for society-wide solutions to problems related to genetic determinism.
When to Not to Use: Do not apply this code when a proposition is rejected based on the identity of the user. Do not apply this code when users agree with each other without justifying their views by refering to a concept related to the worldview of genetic determinism. Do not apply this code when 'the blackpill' is described as a cause of depression or suicidal thoughts.
Example: *No, 'blackpilled' means an understanding that how good looking you are is what will ultimately define your romantic success, no matter how charismatic or sociable you are, particularly as a man. Just because most blackpilled people are misogynistic and believe in patriarchy does not mean these beliefs are necessary to a blackpilled mentality.*

C.4 Self-Hatred

Code: 4

Brief Definition: Criticism or deprecation of ones own self

Full Definition: This refers to the feeling of disdain for the self both based on external appearance and behaviors. Often paired with suicidal feelings or intentions, self-hatred can arise as a comparison of the self to others and end with an individual feeling unhappy of their place in society and the traits they possess.

When to Use: Apply this code when a user uses self-deprecating language to refer to themselves. Apply this code when an incel refers to themselves as 'subhuman.' Apply this code both when an incel shares that they hate their appearance (this can manifest itself as both a low 'rating' of their appearance, or a description of the poor characteristics they possess) Apply this code when an incel shares that they hate the way they act (i.e. a hatred of their mental illness or social skills)

When to Not to Use: Do not apply this code purely on the bases that an incel 'hates their life' or 'wants to rope.' While these motifs are common and indicate an incel's hatred of their situation, it does not necessarily indicate self-hatred. Do not apply this code when an incel indicates that they hate 'society' or 'women. '

Example: *I can say that my life is not going to get better, but to get worse. I am even more subhuman than the subhuman shitskins of my country and to make matters worse I have OCD and diagnosed depression. This is the first time I join a forum, my IQ is so low that I don't know how to write a text clearly.*

C.5 Purity

Code: 5

Brief Definition: Descriptions of what constitues a true incel which can lead into the fear of non-incels infliltrating the incel forum site and incel community

Full Definition: Purity refers to the in-group incel identity and the actual beliefs, traits, and practices of the current members of the in-group staying homogenous and untainted. If out-group memebers are revealed to be in the site, panic within the site can arise due to the memebers of the outgroup being seen as lesser.

When to Use: Apply this code when the post discusses non-incels entering and infiltrating site. This code can also relate to a discussion of whether pre-existing users are true incels and who qualifies as a true incel. This code can also be applied if their is discussion of whom in the site is a fake incel and what the qualities of fake incel are.

When to Not to Use: Do not apply this code if the post generally discusses normal people or "normies" not in relation to the website. Do not use this code if the individual talks of themselves pretending to fit into society or "larping".

Example: *He would most likely have his post deleted and get a considerable warning for bragging. Meanwhile, a Chad like that either putting 0 effort and utterly wasting his looks or pretending to be an incel to mock us would be accepted.*

D Python Script

[2]:
```
import re
import numpy as np
import pandas as pd
from pprint import pprint
import gensim
import gensim.corpora as corpora
from gensim.models import CoherenceModel
from gensim.utils import simple_preprocess
from gensim.models.ldamodel import LdaModel
from gensim.models import Word2Vec, KeyedVectors
import nltk
from nltk.corpus import stopwords
import requests
from bs4 import BeautifulSoup
import pandas as pd
import re
from tqdm import tqdm
from wordcloud import WordCloud
import matplotlib.pyplot as plt
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
import nltk
from nltk.corpus import stopwords
from sklearn.linear_model import LogisticRegression
from nltk.stem import PorterStemmer
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize, sent tokenize
from nltk.stem import SnowballStemmer
from sklearn.metrics import accuracy_score
import concurrent.futures
import random
from retrying import retry
from nltk.corpus import brown
import statsmodels.formula.api as smf
import warnings
warnings.filterwarnings(action='ignore')
```

D.1 Scraping

D.1.1 Create URLs for all pages

```
[ ]: base_url = 'https://incels.is/forums/inceldom-discussion.2/page-'
page_numbers = range(2901, 2973)
urls = [base_url + str(page) for page in page_numbers]
```

D.1.2 Create Soup objects in a list

```
[ ]: # Define a function to retrieve a soup object for a single URL
@retry(stop_max_attempt_number = 3, wait_fixed = 1000)
def get_soup(url):
    response = requests.get(url)
    soup = BeautifulSoup(response.content, 'html.parser')
    return soup

# Use a ThreadPoolExecutor to retrieve soup objects for all URLs in␣
 ↪parallel
with concurrent.futures.ThreadPoolExecutor(max_workers = 10) as executor:
    soup_list = list(tqdm(executor.map(get_soup, urls), total =␣
 ↪len(urls)))

# Print the length of the soup list to confirm that we have 100 soup␣
 ↪objects
print(len(soup_list))
```

D.1.3 Scrape the first layer of Incels

```
[ ]: # creating the lists we need
replies_all = []
views_all = []
text_content = []
links = []
topics = ["Blackpill", "LifeFuel", "SuicideFuel", "RageFuel", "Serious",␣
 ↪"Experiment", "JFL",
          "Venting", "NSFW", "News", "Hypocrisy", "LDAR", "Story", "It's␣
 ↪Over", "TeeHee", "Soy",
          "Cope", "Discussion", "Toxic Femininity", "[Whitepill]",␣
 ↪"Theory", "Brutal", "Based"]
themes_list = []
time_dates = []

for soup in tqdm(soup_list):
```

```
    # Getting replies and views
    dd_tags = soup.find_all('dd')
    replies = []
    views = []
    for i, tag in enumerate(dd_tags):
        content = tag.string.strip()
        if i % 2 == 0 and i < len(dd_tags) - 3:
            replies.append(content)
        elif i % 2 == 1 and i < len(dd_tags) - 3:
            views.append(content)

    # URL-Links and Text Data
    a_tags = soup.find_all('a', {'data-xf-init': True, 'data-preview-url':
↪ True})

    # Iterate through each <a> tag and extract the text content and link
    for a_tag in a_tags:
        text = a_tag.text.strip()
        link = "https://incels.is" + a_tag.get('href')
        text_content.append(text)
        links.append(link)

    # Topics
    cells = soup.find_all('div', {'class': 'structItem-title'})
    themes = []
    for code in cells:
        found_topic = False
        for topic in topics:
            if topic.lower() in code.text.lower():
                themes.append(topic)
                found_topic = True
                break
        if not found_topic:
            themes.append("no topic")

    # Time
    time_tags = soup.find_all('time', {'class': 'structItem-latestDate'})
    for tag in time_tags:
        time_dates.append(tag['datetime'])

    # Append to the lists
    replies_all.extend(replies)
    views_all.extend(views)
```

```
    themes_list.append(themes)

themes_list_flat = [theme for sublist in themes_list for theme in sublist]

# Print the replies and views lists
print("Lengths:", len(replies_all), len(views_all), len(text_content),␣
 ↪len(links), len(themes_list_flat),
      len(time_dates))

# Making sure the lenght of the all columns is the same
target_length = len(text_content)
num_zeros = target_length - len(time_dates)
for i in range(num_zeros):
    time_dates.append(0)
print(len(time_dates))
```

D.1.4 Scrape second layer of Incels.IS

[ ]:
```
# Use a ThreadPoolExecutor to retrieve soup objects for all links in␣
 ↪parallel
with concurrent.futures.ThreadPoolExecutor(max_workers = 10) as executor:
    soup_list_links = list(tqdm(executor.map(get_soup, links), total =␣
 ↪len(links)))

# Print the length of the soup list to confirm that we have the same␣
 ↪number as links
print(len(soup_list_links))

# creating the lists we need
text_posts = []
username = []
response_posts = []
usernames_comments = []

# Loop through each soup object in the list soup_list
for soup in tqdm(soup_list_links):

    # Actual text of the post
    div_tag = soup.find('div', {'class': 'bbWrapper'})
    if div_tag is not None:
        text = div_tag.text.strip()
        text_posts.append(text)
    else:
        text_posts.append('')
```

```
    # Username of the post
    username_span = soup.find('span', {'class': 'username'})
    if username_span is not None:
        username_text = username_span.text
        username.append(username_text)

    # Text of responses
    div_tags = soup.find_all('div', {'class': 'bbWrapper'})
    text = ''
    for div_tag in div_tags[1:]:
        if div_tag is not None:
            text += div_tag.text.strip()
    response_posts.append(text)

    # Usernames of responses
    div_tags_2 = soup.find_all('span', {'class': 'username'})
    usernames = []
    for div_tag in div_tags_2[2:]:
        if div_tag is not None:
            usernames.append(div_tag.text.strip())
    usernames_str = ', '.join(usernames)
    usernames_comments.append(usernames_str)

# Print the length of the text_posts list
print(len(text_posts), len(username), len(response_posts),␣
 ↪len(usernames_comments))

# Making sure the lenght of the all columns is the same
target_length = len(text_content)
num_NaN = target_length - len(username)
for i in range(num_NaN):
    username.append('NaN')
print(len(username))
```

D.1.5 Create the Dataframe and Save as CSV

[ ]:
```
# Combine the three lists into a dictionary
data = {'Replies': replies_all, 'Views': views_all, 'Headline':␣
 ↪text_content, "Topics": themes_list_flat,
        "Date": time_dates, "URLS": links, "Text": text_posts, "Comments":
 ↪ response_posts,
       "usernames_comments": usernames_comments}
```

```
# Create a dataframe from the dictionary
df = pd.DataFrame(data)

# Save the dataframe as a csv file
df.to_csv('[PAGENUMBERS].Incels.IS.csv', index = False)
```

D.2 LDA

D.2.1 Preprocess the Data

[ ]:
```
df = pd.read_csv('incels_data_noduplicates_nonanvaluestext.csv')
```

[7]:
```
def preprocess_data(documents):
 stop_words = stopwords.words('english')

 # Tokenize and remove stopwords
 texts = [[word for word in simple_preprocess(str(doc)) if word not in␣
 ↪stop_words] for doc in documents]

 return texts
```

[8]:
```
posts = df['Text'] ##no comments
```

[9]:
```
processed_texts = preprocess_data(posts)
```

D.2.2 Create a dictionary and a corpus

[9]:
```
# Create Dictionary
id2word = corpora.Dictionary(processed_texts)
```

[10]:
```
# Create Corpus
texts = processed_texts
```

[11]:
```
# Term Document Frequency
corpus = [id2word.doc2bow(text) for text in texts]
```

D.2.3 Run the LDA model

[12]:
```
# Set number of topics
num_topics = 20
```

[13]:
```
# Build LDA model
lda_model = LdaModel(corpus=corpus,
                     id2word=id2word,
```

```
                        num_topics=num_topics,
                        random_state=42,
                        passes=10,
                        alpha='auto',
                        per_word_topics=True)
```

[14]:
```
# Define the number of words you want to display for each topic
num_words = 20

# Display the topics with the specified number of words
topics = lda_model.show_topics(num_topics=num_topics,␣
 ↪num_words=num_words, formatted=False)

for topic in topics:
    print(f"Topic {topic[0]}: {[word[0] for word in topic[1]]}")
```

D.2.4 Evaluate the Model

[16]:
```
coherence_model_lda = CoherenceModel(model=lda_model,␣
 ↪texts=processed_texts, dictionary=id2word, coherence='c_v')
coherence_lda = coherence_model_lda.get_coherence()
print('Coherence Score: ', coherence_lda)
```

```
Coherence Score:  0.5579624667712489
```

D.3 Word2Vec

D.3.1 Load Data

[8]:
```
class MyDataLoader(object):
    """
    A DataLoader class for reading and iterating over a corpus file.

    Args:
        filename (str): The name of the corpus file.
    """
    # initialize the corpus object for a given filename
    def __init__(self, filename):
        self.corpus = filename

    # we will need to define what counts as a "chunk" in this file, so␣
 ↪when the
    # Dataloader is loading (iterating over) the file and feeding it to␣
 ↪the embedding
```

```
    # model, it knows what to treat as one unit. Here, we (arbitrarily)␣
↪say that one
    # line in the file (corresponding to a paragraph) is one chunk.

    def __iter__(self):
    # _iterator function to iterate over the lines of the corpus file.
        for line in open(self.corpus, "r", encoding="utf-8"):
            # checking that the line is not empty:
            if line.strip():
            # you may do some pre-processing on-the-fly. here we tokenize␣
↪and lowercase
            # the string before yielding it
                line = word_tokenize(line)
                line = [x.lower() for x in line]
                yield line
```

[9]:
```
incel_word = MyDataLoader("incel_corpus_preprocesses.txt")
```

D.3.2 Skip-Gram Model Training

[11]:
```
# training the Skip-Gram - model
incel2vec_sg = gensim.models.Word2Vec(
    incel_word,   # the larger corpus object we've loaded
    vector_size=300, # the dimensionality of the target vectors
    window=3,     # window ngram size
    min_count=5,  # ignoring low-frequency words
    epochs=3,      # how many training passes to have
    sg = 1)       # 1 for skip-gram model, 0 for cbow
```

D.3.3 Keyword Extraction

[18]:
```
word_lists = [
    ['foid', 'chad', 'normie', 'fakecel', 'truecel', 'larp'], ## Othering
    ['genetics', 'bully', 'virgin', 'autism', 'gynocentrism',␣
↪'humiliate', 'smv', 'system'], ## Victimhood
    ['blackpill', 'bluepill', 'redpill', 'truth', 'delusional'], ##Truth
]

excluded_words = set(word for sublist in word_lists for word in sublist)


def most_similar_average(words, model, topn=10):
    # Check if the model is of type Word2Vec and get KeyedVectors
```

```
    if isinstance(model, Word2Vec):
        model = model.wv

    # Check if all words are in the model's vocabulary
    valid_words = [word for word in words if word in model]

    if not valid_words:
        raise ValueError("None of the provided words are in the model's␣
 ↪vocabulary.")

    # Get the vectors for valid words
    word_vectors = [model[word] for word in valid_words]

    # Compute the average vector
    average_vector = sum(word_vectors) / len(word_vectors)

    # Find the most similar words to the average vector
    similar_words = model.similar_by_vector(average_vector, topn=topn)

    return similar_words

def filter_similar_words(similar_words, excluded_words):
    return [word for word in similar_words if word not in excluded_words]

# Loop through each list and find the most similar words
for i, words in enumerate(word_lists):
    try:
        similar_words = most_similar_average(words, incel2vec_sg)
        filtered_similar_words = filter_similar_words(similar_words,␣
 ↪excluded_words)
        print(f"List {i+1} similar words: {filtered_similar_words}")
    except ValueError as e:
        print(f"List {i+1} error: {e}")
```

[20]:
```
## Othering
for word_of_interest in ['foid', 'chad', 'normie', 'fakecel', 'truecel',␣
 ↪'larp']:
    use_count = incel2vec_sg.wv.get_vecattr(word_of_interest, "count")
    print(f"'{word_of_interest}' was mentioned {use_count} time(s) in the␣
 ↪corpus")
    print(f"The words most similar to {word_of_interest} are:")
    print(incel2vec_sg.wv.most_similar(word_of_interest))
```

[21]:
```
## Victimhood
for word_of_interest in ['genetics', 'bully', 'virgin', 'autism',␣
↪'gynocentrism', 'humiliate', 'smv', 'system']:
    use_count = incel2vec_sg.wv.get_vecattr(word_of_interest, "count")
    print(f"'{word_of_interest}' was mentioned {use_count} time(s) in the␣
↪corpus")
    print(f"The words most similar to {word_of_interest} are:")
    print(incel2vec_sg.wv.most_similar(word_of_interest))
```

[22]:
```
## Truth
for word_of_interest in ['blackpill', 'bluepill', 'redpill', 'truth',␣
↪'delusional']:
    use_count = incel2vec_sg.wv.get_vecattr(word_of_interest, "count")
    print(f"'{word_of_interest}' was mentioned {use_count} time(s) in the␣
↪corpus")
    print(f"The words most similar to {word_of_interest} are:")
    print(incel2vec_sg.wv.most_similar(word_of_interest))
```

D.4 Extract Close Reading Sample

[122]:
```
othering_words = [
    'foid', 'femoid', 'foids', 'female', 'woman', 'roastie', 'girl',␣
↪'femoids', 'chick',
    'chad', 'chads', 'gigachad', 'chadlite', 'incelchad', 'tyrone',␣
↪'stacy', 'latrel', 'overstacey', 'slops',
    'normie', 'normalfag', 'normies', 'norman', 'nomrie', 'normiesoy',␣
↪'normiewølf', 'normiediocel',
    'fakecel', 'fakecels', 'volcel', 'larper', 'mentalcel', 'truecel',␣
↪'noalcohol', 'rper', 'infiltrator',
    'truecels', 'incel', 'khhv', 'trucels', 'tkhhhv', 'traittruecel',␣
↪'trueceli',
    'larp', 'larping', 'troll', 'larps', 'larpcopexodius', 'trolling',␣
↪'brigader'
]

victim_words = [
    'genetics', 'genes', 'genetic', 'enviourment', 'geneticstldr',␣
↪'genetically', 'looksyour','themnonsense', 'englandgenetics',
    'bully', 'bullies', 'bullying', 'bulllies', 'mock', 'bullied',␣
↪'guessingzombie', 'insulty', 'bullyasiacel', 'losersguys',
    'virgin', 'virgins', 'kissless', 'handholdess', 'khhv', 'danceless',␣
↪'handheldless', 'lookless', 'hugless', 'khhvm',
```

```
    'autism', 'asperger', 'aspergers', 'autistic', 'adhd', 'diagnosed',␣
 ↪'hfa', 'auitism', 'schizophrenia', 'symptomology',
    'gynocentrism', 'hypergamy', 'feminism', 'eastoid', 'disposibility',␣
 ↪'headquaters', 'lovedmainländer', 'manocentrism',
    'widly', 'gynocentric', 'humiliate', 'insulte', 'mock', 'suzuka',␣
 ↪'bullly', 'mandatorly', 'himvisit', 'proceeed',
    'embarrass', 'botit','smv', 'smvslavcel', 'attractivess', 'smvjust',␣
 ↪'svm', 'smvif', 'smva', 'smvi', 'artifically', 'smvit',
    'system', 'systems', 'systemi', 'indiaincline', 'obsolte',␣
 ↪'strenghten', 'bullyimg', 'necessariting'
]

truth_words = [
    'blackpill', 'blackpills', 'bluepill', 'pill', 'blackpillthey',␣
 ↪'blackpilled', 'racepill', 'blackpilli', 'pillpill',
    'bluepilled', 'redpill', 'bluepills', 'redpilll', 'bluebar',␣
 ↪'prototrannies', 'wholheatedly',
    'redpill', 'redpilled', 'redpillers', 'muuuuuuuh',
    'truth', 'truths', 'unvarnished', 'facts', 'reality', 'truthi',␣
 ↪'truthyes', 'factsbecause', 'waifupill',
    'delusional', 'deluded', 'cutehunter', 'retarded', 'incelscopetv',␣
 ↪'articlejfl', 'thintard', 'delusion', 'delutional'
]
```

[11]:
```
# List to store the indices where the word is found
othering_indices = []
victim_indices = []
truth_indices = []

# List to store the count of the words found
othering_count = []
victim_count = []
truth_count = []
purity_count = []

# Filter by
for idx, sublist in enumerate(processed_texts):
    count = 0
    for term in othering_words:
        if term in sublist:
            count += 1
    if count > 0:
        othering_indices.append(idx)
```

```
        othering_count.append(count)

for idx, sublist in enumerate(processed_texts):
    count = 0
    for term in victim_words:
        if term in sublist:
            count += 1
    if count > 0:
        victim_indices.append(idx)
        victim_count.append(count)

for idx, sublist in enumerate(processed_texts):
    count = 0
    for term in truth_words:
        if term in sublist:
            count += 1
    if count > 0:
        truth_indices.append(idx)
        truth_count.append(count)
```

[12]:
```
othering_sample = df.iloc[othering_indices].copy()
victim_sample = df.iloc[victim_indices].copy()
truth_sample = df.iloc[truth_indices].copy()

othering_sample['Counts'] = othering_count
victim_sample['Counts'] = victim_count
truth_sample['Counts'] = truth_count
```

[90]:
```
othering_sample = df.iloc[othering_indices].copy()
victim_sample = df.iloc[victim_indices].copy()
truth_sample = df.iloc[truth_indices].copy()

othering_sample['Counts'] = othering_count
victim_sample['Counts'] = victim_count
truth_sample['Counts'] = truth_count
```

[132]:
```
othering_random_sample = othering_sample.sample(n=40, random_state=42)
victim_random_sample = victim_sample.sample(n=10, random_state=42)
truth_random_sample = truth_sample.sample(n=40, random_state=42)

othering_random_sample.to_csv('othering_random_sample.csv', index=False)
victim_random_sample.to_csv('victim_random_sample.csv', index=False)
truth_random_sample.to_csv('truth_random_sample.csv', index=False)
```

[133]:
```
othering_random_sample = othering_sample.sample(n=20, random_state=42)
victim_random_sample = victim_sample.sample(n=20, random_state=42)
truth_random_sample = truth_sample.sample(n=20, random_state=42)

othering_random_sample.to_csv('othering_test_set.csv', index=False)
victim_random_sample.to_csv('victim_test_set.csv', index=False)
truth_random_sample.to_csv('truth_test_set.csv', index=False)
```

D.5 Classification

D.5.1 Load Data

[3]:
```
df = pd.read_csv('incels_withpreprocessedtext.csv')
def preprocess_data(documents):
 stop_words = stopwords.words('english')

 texts = [[word for word in simple_preprocess(str(doc)) if word not in␣
 ↪stop_words] for doc in documents]

 return texts

posts = df['Text']

processed_texts = preprocess_data(posts)
```

D.5.2 Code Classifier

[4]:
```
themes = {
    "1": {
        "foid": 1,
        "chad": 1,
        "chads": 1,
        "stacy": 1,
        "chadlite": 0.5,
        "femoid": 0.5
    },
    "2": {
        "ugly": 0.5,
        "cope": 0.5,
        "privileged": 0.75,
        "standards": 0.5,
        "feminized": 1,
        "ignore": 1,
        "oppressed": 1
```

```
    },
    "3": {
        "blackpill": 1,
        "blackpilled": 1,
        "bluepill": 1,
        "bluepilled": 1,
        "redpill": 0.5,
        "redpilled": 0.5,
        "antinatalism": 0.5
    },
    "4": {
        "subhuman": 1,
        "hate": 1,
        "flawed": 0.5,
        "suicide": 1
    },
    "5": {
        "truecel": 1,
        "normie": 0.5,
        "larping": 0.75,
        "infiltrator": 1,
        "trolling": 0.75,
        "fakecel": 1,
    }
}
```

[5]:
```
# Define the threshold
threshold = 1.0

# Function to c the theme scores for a thread
def compute_scores(thread, themes):
    scores = {theme: 0 for theme in themes}
    for word in thread:
        for theme, words in themes.items():
            if word in words:
                scores[theme] += words[word]
    return scores

#SINGLE
def classify_thread(thread, themes, threshold):
    scores = compute_scores(thread, themes)
    max_score = max(scores.values())
    if max_score < threshold:
        return "*"
```

```
    return max(scores, key=scores.get)

# Classify all threads in pre-processed text
classified_threads = [classify_thread(thread, themes, threshold) for␣
 ↪thread in processed_texts]
```

[6]:
```
## Count occurance of themes
theme_counts = {theme: 0 for theme in themes}
theme_counts["*"] = 0
for theme in classified_threads:
    theme_counts[theme] += 1
theme_counts
```

[6]: {'1': 48931, '2': 7645, '3': 10767, '4': 15864, '5': 2899, '*': 182817}

[7]:
```
# Calculate relative frequencies
total_threads = len(processed_texts)
relative_frequencies = {theme: count / total_threads for theme, count in␣
 ↪theme_counts.items()}

print("Relative Frequencies of Themes:")
for theme, frequency in relative_frequencies.items():
    print(f"{theme}: {frequency:.2%}")
```

```
Relative Frequencies of Themes:
1: 18.20%
2: 2.84%
3: 4.00%
4: 5.90%
5: 1.08%
*: 67.98%
```

D.5.3 Validate Classifier

[11]:
```
hand_coded_labels = pd.read_csv('closed_coding_test_set.csv')
def preprocess_data(documents):
 stop_words = stopwords.words('english')

 texts = [[word for word in simple_preprocess(str(doc)) if word not in␣
 ↪stop_words] for doc in documents]

 return texts

posts = hand_coded_labels['Text']
```

```
processed_hand_coded_labels = preprocess_data(posts)

rename_dict = {'code_stinus': 'hand_code',}
hand_coded_labels.rename(columns=rename_dict, inplace=True)

# Manually annotated test labels (hand-coded labels)
classified_threads = [classify_thread(thread, themes, threshold) for␣
 ↪thread in processed_hand_coded_labels]
hand_coded_labels['classified_code'] = classified_threads

## Function to calculate the error rate of our classifier - ONE CATEGORY
def classifier_error(column1, column2):

    # Ensure both columns have the same length
    if len(column1) != len(column2):
        raise ValueError("Columns must have the same length")

    # Initialize counters for same and different values
    same_count = 0
    diff_count = 0
    total_count = 0

    # Iterate through pairs of values from both columns
    for val1, val2 in zip(column1, column2):
        # Increment the total count for each pair
        total_count += 1
        # If the values are the same, increment the same_count
        if val1 == val2:
            same_count += 1
        # If the values are different, increment the diff_count
        else:
            diff_count += 1

    # Calculate the percentage of same values
    percentage_same = (same_count / total_count) * 100
    # Calculate the percentage of different values
    percentage_diff = 100 - percentage_same
    return percentage_diff

# Get the columns of hand-coded labels and classified results
column1 = hand_coded_labels['hand_code']
column2 = hand_coded_labels['classified_code']

percentage = classifier_error(column1, column2)
```

```
# Print the error rates
print(f"Error Rate: {percentage:.3f}%")
```

```
Error Rate: 56.667%
```

D.6 Linear Regression

D.6.1 Clean Data

[26]:
```
df = pd.read_csv('incels_data_noduplicates_nonanvaluestext.csv')
df['Classifier'] = classified_threads
```

[210]:
```
# Function to convert 'K' values to numeric values (e.g. '1K' to '1000')
def convert_value(val):
    try:
        # Check if 'K' is in the value and multiply by 1000 if it is
        if 'K' in val:
            return int(val.replace('K', '')) * 1000
        else:
            return int(val)
    except ValueError:
        # Return None for invalid values
        return None

# Apply the function to the DataFrame column
df['Views'] = df['Views'].apply(convert_value)

# Drop rows with None values (optional)
df = df.dropna().reset_index(drop=True)
```

[212]:
```
# Keep only 'Views', 'Replies', and 'Classifier' columns
df_cleaned = df.filter(like='Views').join(df.filter(like='Replies')).
 ↪join(df.filter(like='Classifier'))

# Drop rows where 'Views' is null
df_cleaned = df_cleaned.dropna(subset=['Views'])

# Ensure 'Views' and 'Replies' columns are numeric
columns_to_check = ['Views', 'Replies']
df_cleaned[columns_to_check] = df_cleaned[columns_to_check].apply(pd.
 ↪to_numeric, errors='coerce')

# Drop rows where 'Views' and 'Comment Count' have null values
```

```
df_cleaned = df_cleaned.dropna(subset=columns_to_check).
 ↪reset_index(drop=True)

# Rename Classifier Variables
df_cleaned['Classifier'] = df_cleaned['Classifier'].replace('1',␣
 ↪'Othering')
df_cleaned['Classifier'] = df_cleaned['Classifier'].replace('2',␣
 ↪'Victimhood')
df_cleaned['Classifier'] = df_cleaned['Classifier'].replace('3', 'Truth')
df_cleaned['Classifier'] = df_cleaned['Classifier'].replace('4',␣
 ↪'Selfhate')
df_cleaned['Classifier'] = df_cleaned['Classifier'].replace('5', 'Purity')
df_cleaned['Classifier'] = df_cleaned['Classifier'].replace('*',␣
 ↪'Residual')
```

D.6.2 Regress Number of Views on Topics

[214]:
```
model_views = smf.ols('Views ~ C(Classifier,␣
 ↪Treatment(reference="Residual"))', data=df_cleaned).fit()
```

D.6.3 Regress Replies on Topics

[218]:
```
model_replies = smf.ols('Replies ~ C(Classifier,␣
 ↪Treatment(reference="Residual"))', data=df_cleaned).fit()
```